

## 2 Goal and Area of Application

FAIR data assessment for experimental data is an essential process that ensures data is managed in accordance with the FAIR (Findability, Accessibility, Interoperability, and Reusability) principles. By adhering to these principles, researchers can significantly enhance the transparency, efficiency, and impact of their work beyond the project. Ultimately, a FAIR compliant data resource not only promotes collaboration and innovation but also ensures compliance with institutional and funding agency requirements, ultimately advancing the scientific community's collective knowledge.

This SOP describes a potential approach to evaluate the FAIRness level of the datasets and proposes opportunities for data dissemination into public repositories.

## 3 Definitions and Abbreviations

|          |                                                   |
|----------|---------------------------------------------------|
| FAIR     | Findable, Accessible, Interoperable, and Reusable |
| FAIR-DSM | FAIRplus DataSet Maturity Model                   |

|          |                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metadata | Data that provides information about other data (Source: <a href="https://en.wikipedia.org/wiki/Metadata">https://en.wikipedia.org/wiki/Metadata</a> ). |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4 Method

### 4.1 General

#### Assessment of FAIRness level of dataset

The principles of FAIR have been applied across multiple domains, including life sciences and drug discovery. Given the complex requirements for FAIR data and its associated metadata, a wide array of tools has been developed to assist, guide, and educate those who are not experts in FAIR principles. Many of these tools are cataloged on the FAIRassist.org platform (<https://fairassist.org/>).

Among these tools, we highlight the FAIRplus DataSet Maturity Model (FAIR-DSM) (<https://fairplus.github.io/Data-Maturity/>). Developed within the IMI project FAIRplus, this tool enables the assessment of the FAIR level of a dataset. It offers an exhaustive list of tasks to help users enhance the FAIRness of their datasets in three key areas: content, representation, and hosting environment (Figure 1). By following these guidelines, users can systematically improve the quality and usability of their data.

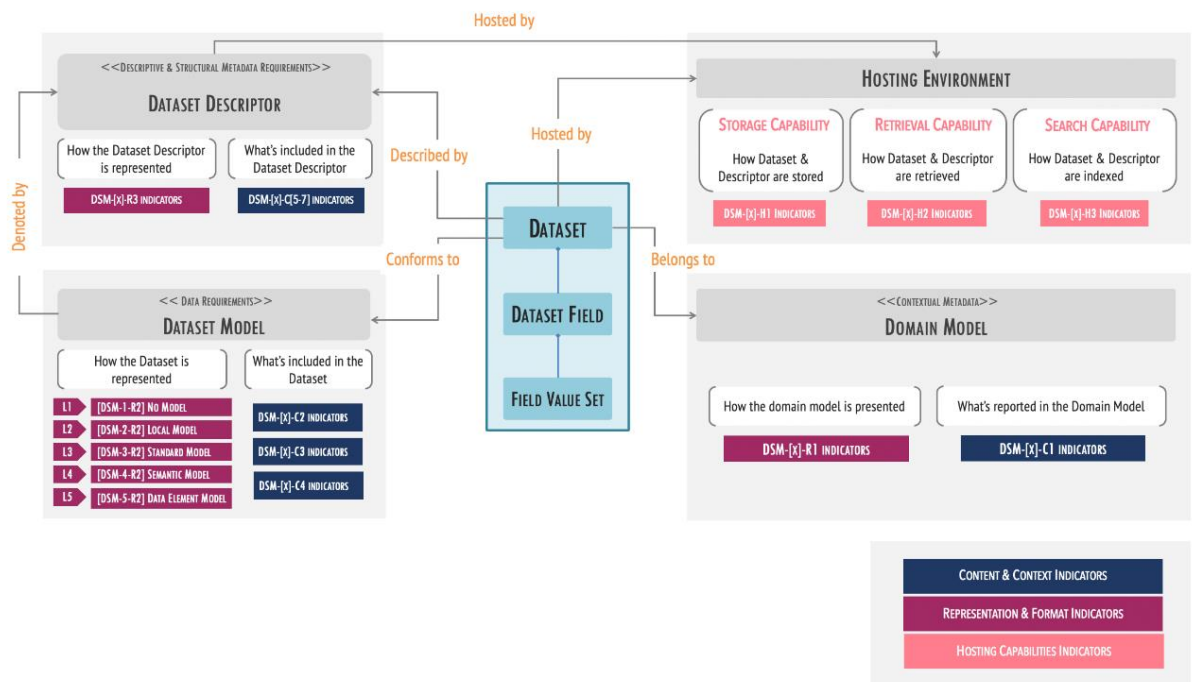


Figure 1: FAIR-DSM domain and infrastructure overview. Figure taken from <https://fairplus.github.io/Data-Maturity/>

Considering the three aforementioned dimensions (i.e. Content and Context, Representation and Format, and Hosting Capabilities), the FAIR-DSM assigns a maturity level to the dataset. This maturity level is a metric indicating the ease of data shareability, ranging from 1 to 5. Level 1 indicates minimal potential for data reuse, while Level 5 represents an enterprise-level standard for sharing and reusing data across communities (Figure 2). This systematic approach helps researchers understand and improve the FAIRness of their datasets, ultimately promoting more effective data management and collaboration.

|   |                         |                                                                                                                                                                                                                                         |
|---|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Managed Data Assets     | <b>Enterprise Level.</b> Data at this level is optimally managed at the most granular level in an environment offering data governance, master data management and reference data management capabilities.                              |
| 4 | Semantically Typed Data | <b>Cross-community Level.</b> This level focuses on cross-domain interoperability and is meant to be the level required for larger harmonization and integration projects.                                                              |
| 3 | Standardised Data       | <b>Community Level.</b> Data at this level complies with community standard domain models, terminologies and formats, and is hosted in an environment offering searching and retrieval capabilities.                                    |
| 2 | Described Data          | <b>Project Level.</b> All datasets generated within a project are consistently described against a locally defined schema, controlled terminologies, and hosted in an environment offering data catalogue level searching capabilities. |
| 1 | Identifiable Data       | <b>Data Object level.</b> Data at this level is identifiable as individual generic data objects and described by generic metadata elements. Hosting environment offers limited retrieval capabilities.                                  |
| 0 | Single Use Data         | No potential for re-use beyond lifetime of the research project                                                                                                                                                                         |

Figure 2: FAIR-DSM maturity levels. Figure taken from <https://fairplus.github.io/Data-Maturity/>

## 4.2 Assessment of FAIRness level of dataset

The process of evaluation of FAIR level of a dataset involves the several key steps:

1. Nomination of a dataset
2. Identification of stakeholders
3. Pre-FAIRification assessment
4. Task force establishment
5. Agile-based tasks sprints
6. Post-FAIRification assessment

### **Nomination of a dataset**

The first step in evaluating a dataset's FAIRness level is to identify and nominate a dataset. In this context, a dataset is a collection of files that represent information in a predefined format and capture data within a similar context. This selected dataset then becomes the basis for the FAIR assessment and subsequent improvement steps. Examples of datasets include an Excel sheet containing experimental endpoint results, such as IC50 values for a compound library, or an analysis report in PDF or Word format from an instrument-based analysis.

### **Identification of stakeholders**

The FAIR-DSM is a questionnaire-based system that enables data owners and analysts to quickly assess the FAIR maturity level of their dataset. The questions cover various aspects of data management, data generation, and legal requirements. Therefore, it is crucial to involve all relevant stakeholders in the evaluation process.

Typically, the stakeholder group includes:

- **Data Managers:** Experts in data handling and knowledgeable about the legal bindings related to the data.
- **Data Owners:** Individuals responsible for generating the data.
- **Data Scientists:** Professionals responsible for downstream analysis of the data.
- **Project Managers or Decision-Makers:** Individuals who can provide the long-term vision for the dataset under evaluation.

By bringing together these diverse stakeholders, the FAIR-DSM ensures a comprehensive evaluation, addressing all facets necessary for improving the dataset's FAIRness.

### **Pre-FAIRification assessment**

Once the required stakeholders have gathered, the next step is to go through the FAIR-DSM questionnaire (<https://fairdsm.biospeak.solutions/>) and provide answers to the best of their knowledge, keeping the selected dataset in context. The questions are in a multiple-choice format, and selecting the most appropriate answers, or all that apply, is recommended.

Since the questions may contain technical and FAIR-specific terms, helper text is provided for each option to aid in understanding (Figure 3). For users interested in deeper insights, an

explanation of how each question relates to the FAIR-DSM model is also available. This comprehensive approach ensures that non-FAIR experts can also use the tool.

The screenshot shows the FAIR-DSM questionnaire interface. At the top, there are three tabs: 'Representation & Format' (selected), 'Content & Context', and 'Hosting Environment Capabilities'. Below the tabs, the focus is on 'Dataset Representation'. A question is displayed: '[2 of 17] Which of the following statements best describe how the shareable Dataset(s) are modeled and represented?'. There are five radio button options:
 

- ☐ Dataset(s) are standardised to a locally defined Dataset Model suitable for data sharing and re-use<sup>[?]</sup>
- ☐ Dataset(s) are standardised to relevant Domain or Community Standard Dataset Model(s) suitable for data sharing and re-use<sup>[?]</sup>
- ☐ Dataset(s) are standardised to a defined Semantic Data Model and represented using Linked Data Representations suitable for data sharing and re-use<sup>[?]</sup>
- ☐ Dataset(s) intended for sharing and re-use are granularly standardised and managed at the Data Element Level (e.g. ISO 11179 MDR standard)<sup>[?]</sup>
- ☐ None of the above

 At the bottom, there are navigation buttons: '< Previous', 'Next >', and a 'Submit' button on the right.

Figure 3: An example question from the FAIR-DSM.

The FAIR-DSM comprises a total of 17 questions. Using smart approaches, only questions relevant to the answers provided will be asked, streamlining the process. Overall, this evaluation takes approximately 30-45 minutes. Upon completion of all relevant questions, the FAIR-DSM generates a report that demonstrates the metric for each FAIR Maturity Level. This metric helps researchers identify the current maturity level of their dataset. For instance, as shown in Figure 4, the current maturity level of the dataset might be Level 0. This report can also be downloaded for internal recording.

| Level   | Representation & Format | Content & Context | Hosting Environment Capabilities | Overall Level % Completion |
|---------|-------------------------|-------------------|----------------------------------|----------------------------|
| Level 1 | 67%                     | 100%              | 100%                             | 86%                        |
| Level 2 | 38%                     | 78%               | 75%                              | 62%                        |
| Level 3 | 0%                      | 0%                | 60%                              | 15%                        |
| Level 4 | 0%                      | 0%                | 67%                              | 14%                        |
| Level 5 | 0%                      | 0%                | 0%                               | 0%                         |

Figure 4: An example of the final high-level report generated by the FAIR-DSM.

Alongside the high-level overview of the current dataset maturity level, the FAIR-DSM provides detailed insights into the criteria that either enable or hinder the dataset's progress towards higher Maturity Levels. For instance, in Figure 5, criteria are color-coded to indicate their status within the dataset: green signifies criteria that are currently satisfied, while red highlights those that are lacking. Achieving all criteria marked in red would elevate the dataset to Maturity Level 1. Furthermore, each criterion is categorized into three dimensions—represented by blue sections—namely, R (representation), C (content), and H (hosting). These dimensions help researchers pinpoint specific areas within the dataset that require attention to enhance its FAIRness level.

Based on this assessment, **2 indicators** still need to be satisfied for your Datasets to reach **Maturity Level 1**

- ☐ [DSM-1-R2] Structured and/or Unstructured Data are organised into Dataset(s) created for the purpose of FAIR sharing
- ☐ [DSM-1-R0] Dataset Metadata is formally represented in the form of an Identifiable Dataset Descriptor
- ☒ [DSM-1-R3] A representation of the Dataset Descriptor conforming to a relevant General Purpose Metadata Schema is available
- ☒ [DSM-1-R1] Contextual Metadata is reported at summary level and represented in the Dataset Descriptor
- ☒ [DSM-1-R4] Dataset Descriptor is available in Machine Readable Format
- ☒ [DSM-1-R5] Dataset(s) available in Machine Readable Format
- ☒ [DSM-1-C0] Each Dataset purposed for FAIR sharing is assigned a unique identifier
- ☒ [DSM-1-C1] Dataset Descriptor includes Descriptive Study/Project-Level summary information
- ☒ [DSM-1-C2] Dataset Descriptor includes Identifying & Descriptive Dataset-Level metadata
- ☒ [DSM-1-C3] Dataset Descriptor contains access information for the Dataset
- ☒ [DSM-1-H1] Metadata hosting environment stores and maintains an identifiable Dataset Descriptor for each identifiable Dataset
- ☒ [DSM-1-H2] The Dataset and its Descriptor are indexed and retrievable (in the same or separate hosting environments) via unique and persistent identifiers
- ☒ [DSM-1-H3] Retrieval of the Dataset and the Dataset Descriptor utilises a standardized communication protocol that is open, free and universally implementable
- ☒ [DSM-1-H4] Metadata hosting environment offers the capability to browse and search contents of the Dataset Descriptor

Figure 5: An example of the maturity level detailed report generated by the FAIR-DSM.

Finally, the pre-FAIR assessment of the dataset is conducted to determine its current maturity level and establish future goals for achieving higher levels of maturity. This assessment also aids in identifying potential technical and non-technical tasks necessary to reach these goals.

### **Task force establishment**

Now that the goals for improving FAIRness have been defined, the next step is to establish a task force responsible for generating and completing tasks aligned with these goals. Typically comprised of data scientists and data engineers, this group possesses expertise in terminologies, ontologies, controlled vocabularies, data models, and web development, among other relevant domains. Their collective knowledge and skills enable them to effectively implement strategies and enhancements aimed at enhancing the FAIRness of the dataset.

### **Agile-based task sprint**

Once the framework for assigning tasks and identifying responsible individuals has been established, it is beneficial to adopt working sprints akin to the Agile methodology (<https://agilemanifesto.org/>). A sprint cycle typically spans 2-3 weeks and involves regular (daily or biweekly) updates from task force members. These sprints serve several purposes:

- a) **Establishing Timeline:** They clarify the timeline for achieving FAIR goals, ensuring transparency and accountability among team members.
- b) **Identifying and Mitigating Blockers:** They facilitate the identification of obstacles hindering progress and prompt actions to resolve these issues swiftly.
- c) **Celebrating Success:** Sprints provide opportunities to acknowledge and celebrate successful completion of tasks, fostering a positive team dynamic and motivation.

This structured approach not only enhances efficiency but also promotes collaboration and ensures alignment with the overarching FAIRness improvement objectives.

#### **Post-FAIRification assessment**

Finally, upon completion of all planned tasks aimed at achieving the FAIR goals, a post-FAIRification assessment using the FAIR-DSM is conducted. This assessment allows the team to systematically evaluate the extent to which the FAIR goals have been achieved and assess the improvement in the dataset's maturity level following task completion. The post-FAIRification assessment serves several key purposes:

- A) It validates whether the set FAIR goals have been met according to the established criteria and standards.
- B) It quantifies and qualitatively evaluates the enhancement in the dataset's FAIRness, comparing it to the baseline determined in the pre-FAIR assessment.
- C) It identifies any remaining gaps or areas requiring further refinement to achieve higher FAIR maturity levels.
- D) Based on the assessment findings, the team can provide feedback on the effectiveness of implemented tasks and iterate on strategies for ongoing FAIR improvement initiatives.

By conducting a thorough post-FAIRification assessment, the team ensures continuous enhancement of data accessibility, interoperability, and usability, thereby supporting more robust and impactful research outcomes.

## **5 Further applicable resources**

The following resources can be referred to understand the importance of FAIR data management and see examples of applicability of FAIR data management on real scenarios.

- <https://fairplus-project.eu/>
- <https://faircookbook.elixir-europe.org/content/home.html>

- <https://doi.org/10.3389/fddsv.2023.1226727>
- <https://doi.org/10.1038/s41597-023-02167-2>

## 6 Document history

*Guidance: Record the version history and reasons for updates.*

| <b>SOP version</b> | <b>Reason</b>                    |
|--------------------|----------------------------------|
| Version 1          | Initial draft of document        |
| Version 2          | <i>E.g., update of section 7</i> |
| Version 3          | <i>E.g., correction of typos</i> |

## 7 Annex

No annex

